Chapter 3

Compositional and Chemical Control of Volatile Polymer-Electrolyte Memristive Dynamics

3.1 Introduction and Scope

?? established, on a single fully characterised *Super Yellow*/*Hybrane*/LiOTf device, that a three-component polymer-electrolyte composite supports the full set of memristive functions in one two-terminal platform. The natural next question — and the subject of this chapter — is how the *dynamics* of these devices respond to deliberate changes in the composite, so that the volatile, fading-memory behaviour can be treated as something to be engineered rather than merely observed.

Three families of measurement recur on the same 2-T geometry throughout this chapter and the next, and are introduced here as a named set. They are (i) current–voltage hysteresis, which reports the steady-state switching window; (ii) potentiation as a function of the number of applied write pulses, which reports how the conductance is built up; and (iii) depotentiation as a function of the delay before read-out, which reports how a written state fades. These three measurements are the shared, abundant basis of the comparative work. The richer single-device synaptic protocols of **??** — excitatory post-synaptic current, separated short- and long-term memory, and spike-timing dependent plasticity — are *not* repeated across the families studied here; where they are invoked, they are used only as prior evidence from the Chapter **??** device and are labelled as such.

Claim discipline. A central feature of this chapter, stated here so that the reader can hold it throughout, is that its results fall into two evidential tiers. The **composition** axis — variation of the ion-transport polymer and salt mass fractions in the *Super Yellow*/PEO/LiOTf system — is supported by a replicated design (typically two to four nominally identical devices per composition cell) and is presented as a *quantitative* result. The **chemistry** axis — variation of the ion-transport host, the salt anion, and the alkali cation — is supported, after data-quality screening, by at most one or two devices per matched cell; it is therefore presented as an *illustrative* landscape, with sample sizes stated explicitly and no claim of statistical power. A third, methodological result establishes that the *drive protocol* itself sets the apparent fading-memory timescale, which both explains why the chemistry comparisons must be read cautiously and provides a design rule for any future comparative study.

3.2 Materials, Devices, and Measurement Protocol

The devices retain the architecture of **??**: a vertical two-terminal sandwich of ITO (bottom, transparent), a solution-processed composite active layer, and a thermally evaporated silver top electrode, with a junction area of $8.25 \times 10^{-2}\,\mathrm{cm}^2$. *Super Yellow* is retained as the semiconducting component in every device of this chapter, so that the electronic-transport role is held fixed and the variation is confined to the ion-transport phase.

Two design axes are varied. The *composition* axis varies the mass fraction of the ion-transport polymer poly(ethylene oxide) (PEO) and of the lithium triflate salt (LiOTf) relative to the semiconductor. The *chemistry* axis substitutes the ion-transport host (PEO or the hyperbranched polyether TMPE), the salt anion (triflate or bis(trifluoromethanesulfonyl)imide, TFSI), and the alkali cation (Li^+, Na^+, K^+, as the triflate or TFSI salt). All devices reported quantitatively use the silver electrode; the small number of gold-electrode devices in the archive are held separate, because the electrode is a confounding variable (section 3.6).

Drive and read protocols. Because the comparisons below turn on them, the electrical protocols are stated explicitly — and, importantly, they are *not* identical across the three measurements (table 3.1). Hysteresis uses a triangular voltage sweep. The potentiation (N-pulse build-up) was acquired with a 3 V write and 1.5 V read at a variable pulse count ($N \leq 1000$); the depotentiation (fading-memory decay) used a fixed train of 30 write pulses at a 4 V write and 2 V read. The inter-pulse interval is effectively common (0.104 s for potentiation, 0.103 s for depotentiation), but the *write amplitude differs* between the two (3 V vs 4 V) — a point that matters when the two are later composed (section 3.8, Chapter 4). The read voltage is above the ionic activation threshold established in **??**,

Table 3.1: The three common measurements use distinct drive protocols (modern *Super Yellow*/PEO/LiOTf/Ag corpus). Crucially, potentiation and depotentiation differ in write amplitude, so the chapter never pools them and Chapter 4 composes them only as an explicit assumption.

Measurement	Write	Read	Pulse train
Hysteresis (HYST)	triangular sweep	—	—
Potentiation (PULSES)	3 V	1.5 V	variable $N \leq 1000$
Depotentiation (DELAYTIME)	4 V	2 V	30 pulses, 0.103 s
Protocol overlay (v114)	3 V, 6 V	1.5 V, 3 V	30 pulses

a point returned to in section 3.6. An older 2022 device generation used a 6 V write with a 3 V read; the modern and old amplitudes are never pooled, and the same-device contrast between two amplitudes is itself the drive-protocol result (section 3.6).

3.3 Methods of Analysis

Hysteresis curves are summarised by the on–off conductance ratio and by a normalised loop area; potentiation curves by the per-decade growth of the conductance ratio with pulse number, by the peak ratio, and by whether the response turns over (decreases) at the highest pulse counts; and depotentiation curves by a fading-memory time constant.

For the depotentiation (fading-memory) decay, the conductance relaxation is described, as in **??**, by a Kohlrausch (stretched-exponential) function

$$\Delta G(t_{\mathrm{wait}}) = \Delta G_0 \, \exp\left[-\left(\frac{t_{\mathrm{wait}}}{\tau}\right)^{\beta}\right] + \Delta G_{\infty}, \tag{3.1}$$

with characteristic relaxation time τ, stretch exponent $0 < \beta \leq 1$, initial change ΔG_0, and residual ΔG_{∞} [1]. Two practical points follow from the data. First, the eight-to-thirteen-point decay curves often under-constrain

β and τ jointly; the fitted τ is therefore reported only where the fit is well identified, and a robust, model-free half-enhancement time $t_{1/2}$ (the delay at which the conductance enhancement has fallen to half its initial value) is used as the primary timescale. Where both are available they agree. Second, fitted β values lie predominantly in the range 0.6–0.9, i.e. genuinely stretched relaxation, consistent with the heterogeneous distribution of activation barriers invoked in **??**.

Data-quality control. Two screening layers are applied before any number is reported, both recorded so that the analysis is reproducible. Individual measurements that are erratic or non-physical are flagged and excluded. In addition, a per-device visual review of the recorded curves was carried out; the resulting verdicts — whether a curve is used as measured, cleaned by removing identified anomalous points, or discarded — together with the exact points retained, are stored in a curation registry and applied automatically by the analysis scripts, rather than being made ad hoc. The fitted timescale obtained directly from the instrument-side pipeline was found not to be robust to such anomalous points, so all timescales reported here are recomputed after this screening.

3.4 Composition Control of the Dynamics

The quantitative core of this chapter is the *Super Yellow*/PEO/LiOTf composition grid, in which the PEO mass fraction is varied over 0.3, 0.6 and 1.2 and the salt mass fraction over 0.045, 0.09 and 0.18, with two to four nominally identical silver-electrode devices per cell. Each measurement was acquired under its own fixed protocol (table 3.1); the composition contrasts below are therefore always drawn *within* a single measurement, never by pooling the 3 V potentiation with the 4 V depotentiation. Three trends, mutually consistent and each visible across the

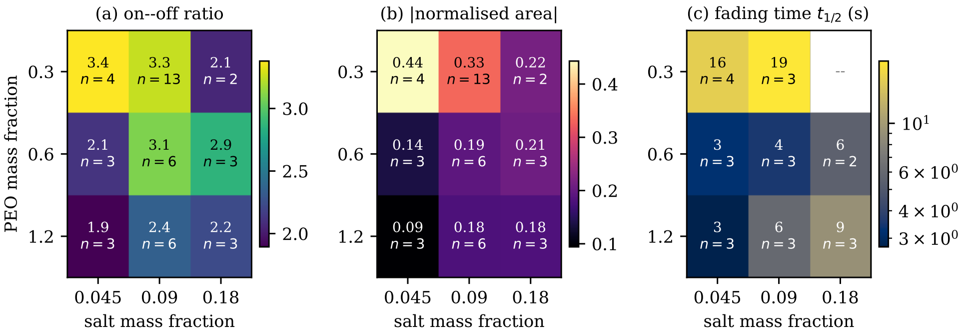

Figure 3.1: Composition control of the steady-state and dynamical response of *Super Yellow*/PEO/LiOTf devices. The switching window (a, b) and the fading-memory time (c) both decrease as the PEO mass fraction increases at fixed salt content.

grid, define the result.

First, the steady-state switching window contracts as the PEO fraction rises. At fixed salt mass fraction 0.09, the median on–off ratio falls from ≈ 3.3 at PEO 0.3 to ≈ 2.4 at PEO 1.2, and the normalised loop area falls correspondingly from ≈ 0.33 to ≈ 0.18; the activation voltage is, by contrast, essentially flat across the whole grid (2.2 V–2.4 V), indicating that composition tunes how *much* the conductance moves rather than the voltage at which it begins to move. The widest window in the grid sits in the low-PEO row, with the largest loop area at the low-salt corner (PEO 0.3, salt 0.045: area ≈ 0.44); the per-sweep on–off ratio is strongly right-skewed there (cell median ≈ 3.4 but individual sweeps reaching ~ 40), so robust cell medians rather than means are reported throughout to avoid letting a few outlying sweeps define a cell.

Second, potentiation weakens, and changes shape, as the composition is varied (fig. 3.2). Two axes act distinctly. The PEO fraction controls the *strength* of pulse integration: the per-decade growth of the conductance ratio (the log–log slope α) falls with PEO content — most cleanly at the lowest salt, where it drops from $\alpha \approx 1.1$ (near-linear accumulation) at PEO 0.3 to $\alpha \approx 0.3$ (strongly compressive) at PEO 1.2 — and the peak conductance ratio, i.e. the usable dynamic range, is largest in the low-PEO, low-salt corner ($\approx 480\times$ at PEO 0.3/salt 0.045) and falls to a few-fold at high PEO. The salt fraction, by contrast, controls *how far* the response can be driven before it saturates: at the lowest salt (0.045) every cell *turns over* — the conductance ratio peaks and then declines — by $N \sim 100$–300 pulses, whereas at the highest salt (0.18) the response grows monotonically through the largest counts tested ($N \approx 1000$), with salt 0.09 intermediate. Two caveats are kept in view: the α trend is noisier at the higher salts, where the sparsely sampled PEO 0.3/salt 0.18 cell ($n = 2$, curation-salvaged) departs from it; and the turnover sets a practical ceiling on the usable pulse count. Read constructively, these responses are a composition-tunable, predominantly compressive and (in the low-salt cells) non-monotone integration of the input pulse train — the nonlinear input encoding that Chapter 4 builds on.

Third, and most important for the rest of the thesis, the fading-memory timescale is composition-tunable over roughly 2 s–20 s. The longest retention occurs in the low-PEO row (at PEO 0.3, $t_{1/2} \approx 19\,\mathrm{s}$ at salt 0.09, with an identified $\tau \approx 25\,\mathrm{s}$); increasing the PEO fraction to 0.6 or 1.2 shortens the timescale to a few seconds. A single curation-salvaged device at the still-lower PEO 0.15 level ($t_{1/2} \approx 22\,\mathrm{s}$) is consistent with this trend, though it lies outside the replicated grid and is not counted toward it. The same ordering is reproduced by the instrument-side single-exponential time constant computed independently, which provides an internal cross-check on the trend. The device-to-device scatter within each cell is substantial and is not treated here as mere noise: the spread of timescales *across* the grid, together with the within-cell variability, constitutes the heterogeneous bank of fading-memory elements that Chapter 4 exploits.

Taken together, these observations support a single physical reading: increasing the proportion of the ion-transport phase produces a weaker,

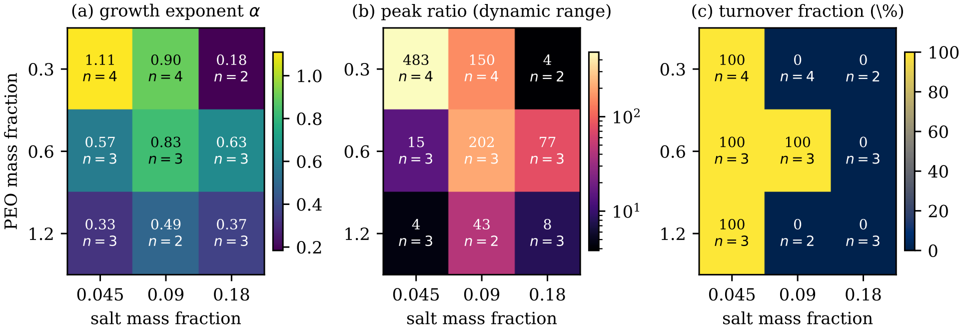

Figure 3.2: Potentiation (pulse-integration) response across the *Super Yellow*/PEO/LiOTf composition grid: (a) the log–log growth exponent α, (b) the peak conductance ratio (usable dynamic range, log scale), and (c) the fraction of devices whose response turns over within the measured range ($N \leq 1000$). Increasing the PEO fraction lowers α and the dynamic range; increasing the salt fraction suppresses turnover, sustaining growth to higher pulse counts. Per-cell median, with the device count n annotated in each cell.

faster-relaxing memristive response, while the salt content governs how far that response can be pushed before it saturates. This is the quantitative spine of the chapter, and the one result that rests on replicated devices.

3.5 Chemical-Tuning Landscape

Beyond composition, the identity of the electrolyte components shifts the dynamics further. These comparisons are reported as an illustrative landscape: each matched cell (fixed composition, electrode, and protocol) contains at most one or two screened devices, so the magnitudes below are indicative and the sample size is given in every case.

Host. Substituting the linear polyether PEO by the hyperbranched polyether TMPE, at fixed lithium-triflate chemistry and composition, shortens the fading-memory time by roughly a factor of six (PEO/LiOTf:

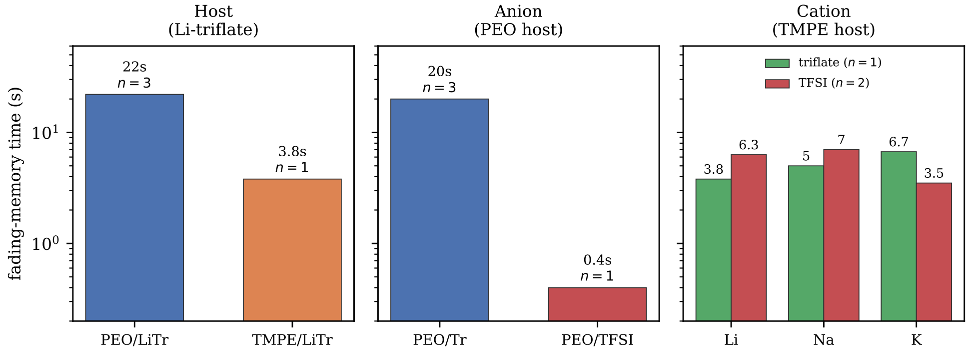

Figure 3.3: Illustrative effect of electrolyte chemistry on the fading-memory time constant (*Super Yellow*, 0.3/0.09, silver, matched 4 V/2 V protocol). Host (PEO vs TMPE, Li-triflate) and anion (triflate vs TFSI, PEO host) produce the largest shifts. The cation panel shows the same three cations in the TMPE host under each anion: the order *inverts* with the anion (potassium longest in triflate, shortest in TFSI), so the cation produces no consistent, host- and anion-independent ordering. Sample sizes are small (n as labelled); see text.

$t_{1/2} \approx 20\,\mathrm{s}$–$25\,\mathrm{s}$, three screened silver devices; TMPE/LiOTf: $\approx 4\,\mathrm{s}$, one screened silver device with a clear decay). This is the best-replicated of the chemistry comparisons and the clearest single chemistry effect, although the TMPE side rests on a single clean device.

Anion. Replacing triflate by the more charge-delocalised TFSI anion, in the PEO host, shortens the fading-memory time dramatically — from $\approx 20\,\mathrm{s}$ (triflate) to of order $1\,\mathrm{s}$ (TFSI). The TFSI decays additionally tend to a sharper, more abrupt (compressed) shape rather than the stretched form of the triflate devices, suggesting a qualitatively different relaxation-time distribution.

Cation. The original motivating hypothesis — that the alkali cation would order the fading-memory timescale as $Li^+ > Na^+ > K^+$, by analogy with the hard–soft acid–base argument of **??** [2] — is *not* supported by the data. The decisive evidence is a pair of cation series measured in the *same* host (TMPE), at the same composition, electrode, and protocol, differing only in the salt anion (the cation panel of fig. 3.3). With triflate, the timescale *increases* along the series — $Li^+ \approx 3.8\,s$, $Na^+ \approx 5.0\,s$, $K^+ \approx 6.7\,s$ (one screened device per cation, all $R^2 \geq 0.93$) — so potassium retains *longest*, the reverse of the hypothesised order. With TFSI, the same three cations invert: potassium is now *shortest* ($\approx 3.5\,s$) and lithium and sodium are together and longer ($\approx 6\,s$–$7\,s$, two devices per cation). A single change of anion thus flips which cation persists longest, while in each series the within-cation device-to-device variation is comparable to the between-cation differences. The honest conclusion is that, in this archive, cation identity does not produce a robust, host- and anion-independent retention ordering: any apparent ordering (such as potassium being shortest in the TFSI family, which is consistent with the weakest cation–oxygen coordination) is contingent on the rest of the chemistry. A properly powered test of the cation hypothesis would require a dedicated series at fixed host, anion, protocol, and electrode with at least three replicates per cell, and is identified as future work in section 3.8.

The hierarchy that emerges from this landscape is therefore that the host and the anion shift the fading-memory time more strongly, and more legibly, than the cation does.

3.6 The Drive Protocol Sets the Timescale

A methodological result underlies, and disciplines, every comparison above. Because the read voltage used to interrogate the state is above the ionic activation threshold, and because the write amplitude controls how far the ionic profile is driven, the *drive protocol* itself influences the apparent fading-memory time. The point is made cleanly by a single device measured under both protocols: its fitted relaxation time increases

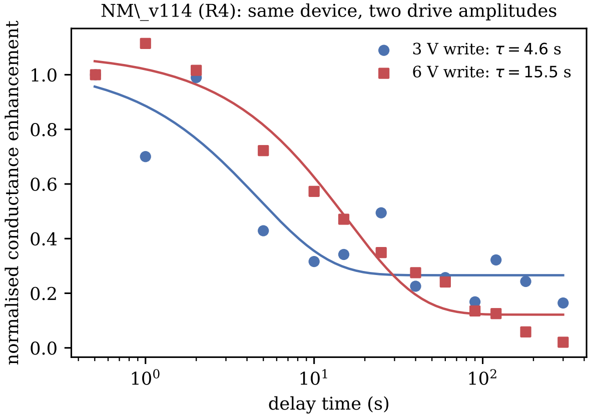

Figure 3.4: The apparent fading-memory time is set in part by the drive protocol. The same device, measured under a larger write/read amplitude, shows a markedly longer apparent relaxation time.

from $\approx 4.6\,\mathrm{s}$ under the 3 V-write/1.5 V-read protocol to $\approx 15.5\,\mathrm{s}$ under the 6 V-write/3 V-read protocol — a roughly three-fold change in the measured timescale with no change in material, only in drive.

The mechanistic reading is that a larger write amplitude displaces more of the ionic charge (and, at the largest amplitudes, may drive electrode-level electrochemical processes), producing a deeper, longer-lived written state, while the supra-threshold read perturbs the state during interrogation. Two consequences follow. First, the composition result of section 3.4 is unaffected, because it was obtained entirely within a single, fixed protocol; only the chemistry comparisons, which unavoidably span device generations, are exposed to this confound, and they are reported accordingly. Second, the result is a positive design rule for the field: the volatile timescale of these devices is not a fixed material constant but is co-determined by how the device is driven, and any meaningful cross-device or cross-chemistry comparison of retention must hold the write/read protocol, the electrode, and the composition fixed.

3.7 Discussion

The picture that emerges is of a device whose volatile dynamics are controlled by a hierarchy of levers. The strongest and most reliable is the *composition* of the ion-transport phase, which tunes the switching window, the potentiation strength, and the fading-memory time together and in the same direction: more ion-transport polymer yields a weaker, faster-relaxing device. Next come the *host* and *anion* of the electrolyte, which shift the fading-memory time over a wider range still but are documented here only illustratively. The *cation*, which the proof-of-concept chapter's mechanism had suggested would be the controlling variable, is in fact the weakest and least legible lever in this dataset, and its effect is not separable from host, anion, and drive protocol at the available sample sizes. The hard–soft acid–base framework of **??** is retained here as a qualitative organising idea — it correctly anticipates, for instance, that the most weakly coordinating cation relaxes fastest within one family — but it is explicitly not advanced as a quantitative, transferable law on the strength of these data.

A working mechanistic picture. The composition trends admit a cautious interpretation, offered here as a working hypothesis rather than a demonstrated microscopic mechanism. Increasing the PEO (ion-transport) fraction dilutes and interrupts the *Super Yellow* electronic-percolation pathway, which plausibly accounts for the simultaneous contraction of the switching window and the weaker pulse-driven conductance modulation; the same ion-richer matrix should also let the displaced ionic profile relax more readily, shortening the fading-memory time — so that window, potentiation strength, and retention move together, as observed. The salt fraction, by contrast, plausibly sets the number of mobile, chargeable ionic states available before saturation, which would explain why a higher salt content sustains potentiation to larger pulse counts before turning over. The host and anion effects are most naturally read in terms of cation–ether coordination strength, ion pairing, and segmental mobility — that is, the breadth and depth of the distribution of local activation barriers that underlies the stretched-exponential relaxation. None of these statements is established microscopically here; impedance spectroscopy and temperature-dependent measurements (section 3.8) would be needed to test them, and the hard–soft acid–base idea is kept strictly qualitative.

The variability that runs through the chapter deserves a positive rather than an apologetic reading. The spread of fading-memory times, both across compositions (roughly an order of magnitude) and between nominally identical devices, is precisely the kind of heterogeneity that a physical reservoir requires: a bank of elements with diverse but individually reproducible-enough time constants. This is made concrete in fig. 3.5, which places every screened silver device in a fading-memory ($t_{1/2}$) versus dynamic-range (peak ratio) plane: the PEO fraction sets the broad region a device occupies — low PEO in the long-memory, high-dynamic-range corner; high PEO in the opposite one — so a single composition selects a *paired* operating point, while the substantial scatter within each composition is the genuine device-to-device diversity. Chapter 4 takes up this point directly.

3.8 Limitations and Future Work

The principal limitation is one of statistical power off the composition axis. Only the *Super Yellow*/PEO/LiOTf composition grid is replicated; the host, anion, and cation comparisons rest on one or two screened devices per matched cell, and are therefore illustrative. Three specific consequences should be stated plainly. First, the cation hypothesis is not tested here at adequate power, and the apparent absence of a $Li^+ > Na^+ > K^+$ ordering should be read as “not demonstrated in this archive” rather than as a

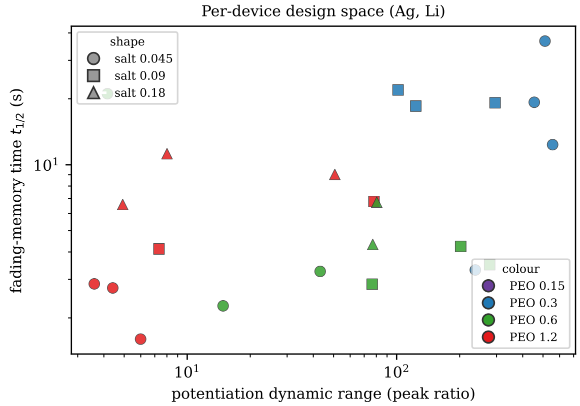

Figure 3.5: Per-device design space of the *Super Yellow*/PEO/LiOTf/Ag composition grid: fading-memory time $t_{1/2}$ against potentiation dynamic range (peak ratio), coloured by PEO fraction and marked by salt fraction (one point per device, median over its pixels). The PEO fraction selects the broad region a device occupies — low PEO (blue) in the long-memory, high-dynamic-range corner, high PEO (red) in the opposite one, so memory and nonlinearity are not independently tunable but *paired*. The substantial within-composition scatter is the genuine device-to-device heterogeneity that Chapter 4 exploits; it is shown, not averaged away.

positive refutation. Second, the supra-threshold read and the amplitude dependence of the apparent timescale (section 3.6) mean that absolute retention values are protocol-specific. Third, electrode (silver versus gold) and ageing effects are additional uncontrolled variables that were held out rather than modelled. Fourth — and most consequential for Chapter 4 — pulse integration and forgetting were characterised *separately and at different write amplitudes*: the potentiation curves were acquired at a 3 V write and the fading-memory decays at a 4 V write (table 3.1), at an effectively common inter-pulse interval, and the archive contains no experiment that varies the input timing to probe the interaction of integration and forgetting directly. Because the drive amplitude itself co-sets the apparent timescale (section 3.6), composing the 3 V potentiation shape with the 4 V decay in Chapter 4 assumes the two regimes are compatible — an explicit modelling assumption rather than a measured fact. The clean test is a single matched-amplitude experiment that sweeps both pulse count and inter-pulse interval; it is named as future work.

The corresponding future-work programme is therefore well defined: a dedicated cation series in a single host and anion, at a single fixed protocol and electrode, with a sub-threshold read voltage (as in the proof-of-concept device of **??**) and at least three replicates per cell, would convert the illustrative chemistry landscape into a powered comparison. Encapsulation and a systematic ageing study would address the stability dimension.

3.9 Conclusions and Bridge to Chapter 4

This chapter has shown that the volatile dynamics of *Super Yellow*/polymer-electrolyte composites are controllable, and that the control is dominated by composition. Quantitatively, the fading-memory time of the *Super Yellow*/PEO/LiOTf system is tunable over roughly 2 s–20 s by composition alone, with the switching window and potentiation strength varying in concert; chemistry (host, anion) extends this range further, illustratively; and the cation, contrary to the initial hypothesis, is not a robust controlling variable at the available power. A methodological result — that the drive protocol co-sets the apparent timescale — frames how all of this must be measured and compared.

What Chapter 4 requires from the present chapter is precisely a set of behavioural ingredients: a composition-indexed family of fading-memory time constants spanning a useful range, a characterisation of how conductance is built up by pulses (including its saturation and turnover), and an honest account of device-to-device heterogeneity. Those ingredients are now in hand. They also define the shape of the design space the next chapter inherits: the fading-memory time and the input nonlinearity (α)

are *coupled* through the PEO fraction (fig. 3.5) — a single composition selects a paired (memory, nonlinearity) operating point rather than two independent settings — while the salt fraction acts as a partially separate lever on the usable dynamic range and turnover. A temporal-computing substrate must therefore be assembled from a *bank* of compositions to span the behaviour it needs, rather than tuned to a single optimum; the heterogeneity catalogued here is what makes that bank possible. The next chapter treats them not as imperfections of a memory but as the raw material of a temporal-computing substrate.

References

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2. Pearson, R. G. Hard and Soft Acids and Bases. *Journal of the American Chemical Society* **85,** 3533–3539. doi:10.1021/ja00905a001 (1963).